CYBER THREAT INTELLIGENCE & ATTACK PATTERN DISCOVERY

Automated Defense Analysis Report | Generated: October 18, 2026 - 10:40:02 UTC | Source: sample_network_traffic.csv

1. Executive Threat Briefing

Analysis of 300 traffic records revealed 142 potentially suspicious flows (47.3% of analyzed traffic) and 26 statistical anomalies. Primary detected threat patterns include: DDoS, PortScan, SSH-Patator, FTP-Patator, DoS-Hulk, Botnet, Web Attack - SQL Injection. Immediate review of exposed edge services and firewall rate-limiting policies is recommended.

Total Analyzed Flows	Baseline Normal Flows	Potentially Suspicious	Statistical Anomalies	Suspicious %
300	**158**	**142**	**26**	**47.3%**

2. Machine Learning Model Specifications & Validated Metrics

Model Architecture	Accuracy	Precision (Macro)	Recall (Macro)	F1-Score (Macro)
Random Forest	0.9982	0.9970	0.9970	0.9970

3. Discovered Attack Patterns & MITRE ATT&CK; Alignment

Detected Threat Pattern	Flow Count	Traffic %	MITRE ID	Technique Name	Tactic
DDoS	38	12.7%	T1498	Network Denial of Service	Impact
PortScan	34	11.3%	T1046	Network Service Discovery	Discovery
SSH-Patator	21	7.0%	T1110.001	Password Guessing (SSH)	Credential Access
FTP-Patator	15	5.0%	T1110.001	Password Guessing (FTP)	Credential Access
DoS-Hulk	15	5.0%	T1499	Endpoint Denial of Service	Impact
Botnet	9	3.0%	T1071.001	Application Layer Protocol (C2)	Command and Control
Web Attack - SQL Injection	9	3.0%	T1190	Exploit Public-Facing Application	Initial Access

4. Academic & Forensic Integrity Notice

LEGAL & FORENSIC DISCLAIMER: *All classifications, confidence scores, and anomaly ratings presented in this report are probabilistic outputs generated by Machine Learning models. Outputs are labeled as 'Model Prediction' and 'Potentially Suspicious'. These results do not represent definitive proof of malicious compromise or intentional intrusion without secondary corroboration through full packet inspection (PCAP), endpoint logs, and qualified cybersecurity analyst triage.*